

Shizhe Hu

(IEEE Senior Member, ACM/CCF/CSIG/CAA Member, Doctoral Supervisor)

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<https://www7.zzu.edu.cn/csai/info/1232/1895.htm> • Google Scholar Profile • GitHub

Biography

I am currently an associate professor affiliated with the School of Computer & Artificial Intelligence at Zhengzhou University (TOP Thirty in China). I obtained the Ph.D degree of Software Engineering from Zhengzhou University in 2021, supervised by Professor Yangdong Ye. I obtained the honors of ACM Rising Star Award, ACM Zhengzhou, 2023 and ACM Outstanding Ph.D. Dissertation Award, ACM Zhengzhou, 2021.

My primary research interests focus on Information Bottleneck Theory, Multi-view/Multi-modal Learning, Trustworthy Analysis, Continual Learning, Granular Computing, Federated Learning, Clustering Analysis, Unsupervised Representation Learning, and Large Model Evaluation and Safety. He has published more than 30 papers on the TOP journals and conferences, such as IEEE TPAMI, TIP, TKDE, TNNLS, TCYB, Information Fusion, Pattern Recognition, ICLR, NeurIPS, ICML, CVPR, AAAI, IJCAI and ACM MM. In addition, [He also served as the Associate Editor \(AE\) of IEEE Transactions on Image Processing \(IEEE TIP\) and Pattern Recognition \(PR\), and Editorial Board member of Information Processing and Management \(IP&M\)](#), and also served as the Area Chair (AC) of ICLR and ICML, and Senior Program Committee (SPC) of AAAI and IJCAI-ECAI. He served as the reviewer of more than 30 international journals and conferences, including IEEE TPAMI, TNNLS, TIP, ICML, NeurIPS, AAAI, IJCAI, CVPR, ACM MM, etc. He is an IEEE Senior Member.

Research Interest

Information Bottleneck Theory, Multi-view/Multi-modal Learning, Trustworthy Analysis, Continual Learning, Granular Computing, Federated Learning, Clustering Analysis, Unsupervised Representation Learning, and Large Model Evaluation and Safety

Education

- **Zhengzhou University** **Zhengzhou, Henan, China**
Software Engineering, Ph.D. 2015 – 2021
Dissertation: Research on Theory and Algorithm of Propagation Information Bottleneck
Adviser: Yangdong Ye
 - **Zhengzhou University** **Zhengzhou, Henan, China**
Computer Science and Technology, B.A. 2011 – 2015
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Professional Experiences

- **Zhengzhou University** **Zhengzhou, Henan, China**
Associate Professor 2024.01 – now
 - **Zhengzhou University** **Zhengzhou, Henan, China**
Assistant Professor 2021.08 – 2024.01
 - Conducted fundamental and applied research in computer vision and deep learning.
 - Theoretical fields: Information bottleneck theory, and multi-view learning theory.
 - Applied fields: Multi-modal learning, unsupervised representation learning and clustering.
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Awards and Honors

- [ACM Rising Star Award](#), ACM Zhengzhou, 2023.
- [ACM Outstanding Ph.D. Dissertation Award](#), ACM Zhengzhou, 2021.
- [First Prize](#) of Outstanding Scientific and Technological Paper of Henan Province for paper "View-wise Versus Cluster-wise Weight: Which Is Better for Multi-view Clustering?", 2024.

- **Second Prize** of Natural Science Award of Henan Province for paper "Dynamic Auto-weighted Multi-view Co-clustering", Henan Province, 2021.
 - **Outstanding Ph.D. Dissertation Award**, Zhengzhou University, 2021.
 - **Excellent Graduate of Henan Province**, Department of Education of Henan Province, 2015, 2021.
 - **Best Paper Award** in the 15th Forum of Postgraduates, Zhengzhou University, 2021.
 - **Outstanding Achievement Award**, Zhengzhou University, 2020.
 - **Outstanding Young Volunteer**, School of Information Engineering of Zhengzhou University, 2016.
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Academic Services

- **Associate Editor**: IEEE Transactions on Image Processing (IEEE TIP) [CCF-A, CAS Rank 1], 2025.12-Now
 - **Associate Editor**: Pattern Recognition [CCF-B, CAS Rank 1], 2026.01-Now
 - **Editorial Board**: Information Processing and Management (IP&M) [CCF-B, CAS Rank 1], 2025.07-Now
 - **Guest Editor**: Information Fusion [CAS Rank T1], 2026.01-Now, Special Issue: Multi-view Fusion and Learning on Low-quality Data: Foundation Models in Theories, Algorithms and Applications
 - **Guest Editor**: Pattern Recognition (PR) [CCF-B, CAS Rank T1], 2025.11-Now, Special Issue: Evolving Multi-View Learning: From Theory to High-Impact Applications
 - **Guest Editor**: Technologies [CAS Rank T3], 2025.11-Now, Special Issue: Advances in the Information Bottleneck: Theory, Methods, and Applications
 - **Area Chair**: International Conference on Learning Representations (ICLR 2026)
 - **Area Chair**: International Conference on Machine Learning (ICML 2026)
 - **Senior Program Committee (SPC)** : International Joint Conference on Artificial Intelligence (IJCAI-ECAI 2026)
 - **Senior Program Committee (SPC)** : AAAI Conference on Artificial Intelligence (AAAI 2027)
 - **Session Chair**: 33rd International Joint Conference on Artificial Intelligence (IJCAI 2024): Session ML "Multi-view Learning"
 - **Session Chair**: 23rd IEEE International Conference on Data Mining (ICDM 2023): Session A3-III "Federated Learning and Ensembles & Security and Privacy"
 - **Workshop Chair**: 1th International Conference on Computer Vision and Deep Learning (CVDL 2024): Workshop on "New Multi-view Learning Models and its Application"
 - **Invited Speaker**: 2022 IEEE 5th International Conference on Pattern Recognition and Artificial Intelligence (PRAI 2022)
 - **Spotlight Paper**: 2024 Chinese Congress on Image and Graphics (CCIG 2024) at Xian, China, 24-May-2024
 - **Reviewer**: IEEE TPAMI, TIP, TKDE, TNNLS, TMM, TCSVT, TGRS, Information Fusion, Pattern Recognition, Neural Networks, Artificial Intelligence Review, Information Processing & Management, Knowledge-Based Systems, International Journal of Machine Learning and Cybernetics, MultiMedia Tools and Applications, Cognitive Computation.
 - **Technical Committee**: ICML, NeurIPS, CVPR, ECCV, ACM MM, AAAI, IJCAI, ICDM, PAKDD, PRCV, CCDM, PRAI
 - **Member**: IEEE/ACM/CCF/CSIG Member
 - **Committee**: CSIG-Big Visual Data, CCF-MM, CSIG-MM, CAA-PRMI
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Fundings

1. Natural Science Foundation of China (**NSFC General Program**), 2026.01-2029.12.
 2. Natural Science Foundation of China (**NSFC Youth Program**), 2023.01-2025.12.
 3. The 17th China Postdoctoral Science Foundation (**Special**), 2024.03-2026.02.
 4. The 74th China Postdoctoral Science Foundation (**General**), 2023.08-2025.03.
 5. Henan Province **Outstanding Youth** Science Fund Program, 2025.01-2027.12.
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Publications-Journal

- **A Survey on Information Bottleneck**
Shizhe Hu, Zhengzheng Lou, Xiaoqiang Yan, and Yangdong Ye, [IEEE TPAMI 2024](#), [Popular Articles in 07/2024](#).
- **Multi-task Image Clustering through Correlation Propagation.**
Shizhe Hu, Xiaoqiang Yan, Yangdong Ye, [IEEE TKDE 2021](#)
- **Granular Information Bottleneck for Deep Multi-modal Clustering**
Zhengzheng Lou, Mingyang Lv, Yuhao Zhan, Yingxuan Li, Chaoyang Zhang, Shizhe Hu (Corresponding author), [IEEE TIP 2026](#).
- **Parameter-free Deep Multi-modal Clustering with Reliable Contrastive Learning**
Zhengzheng Lou, Hang Xue, Yanzheng Wang, Chaoyang Zhang, Xin Yang, Shizhe Hu (Corresponding author), [IEEE TIP 2025](#).
- **Multi-View Clustering via Triplex Information Maximization**
Chaoyang Zhang, Zhengzheng Lou, Qinglei Zhou, and Shizhe Hu (Corresponding author), [IEEE TIP 2023](#)
- **View-wise Versus Cluster-wise Weight: Which Is Better for Multi-view Clustering?**
Shizhe Hu, Zhengzheng Lou, and Yangdong Ye, [IEEE TIP 2022](#)
- **Structure-enhanced Self-supervised Weighted Information Bottleneck for Multiview Clustering**
Zhengzheng Lou, Yucong Wu, Ke Zhang, Chaoyang Zhang, Bo Ji, and Shizhe Hu (Corresponding author), [IEEE TNNLS 2026](#).
- **Deep Multi-view Clustering by Pseudo-label Guided Contrastive Learning and Dual Correlation Learning**
Shizhe Hu, Chengkun Zhang, Guoliang Zou, Zhengzheng Lou, and Yangdong Ye, [IEEE TNNLS 2025](#).
- **Multiview Clustering with Propagating Information Bottleneck**
Shizhe Hu, Zenglin Shi, Xiaoqiang Yan, Zhengzheng Lou, and Yangdong Ye, [IEEE TNNLS 2024](#)
- **DMIB: Dual-correlated Multivariate Information Bottleneck for Multi-view Clustering**
Shizhe Hu, Zenglin Shi, Yangdong Ye, [IEEE TCYB 2022](#)
- **Stop Pushing Friends Away: Trustworthy Hybrid Contrastive Multi-modal Clustering.**
Guoliang Zou, Shizhe Hu* (Co-Corresponding author), Tongji Chen, Yangdong Ye*, [Information Fusion 2026](#)
- **Dynamic Auto-weighted Multi-view Co-clustering**
Shizhe Hu, Xiaoqiang Yan, Yangdong Ye, [Pattern Recognition 2020](#)
- **Contrastive cross-modal clustering with twin network**
Yiqiao Mao, Xiaoqiang Yan, Shizhe Hu, Yangdong Ye. [Pattern Recognition 2024](#)
- **Nice to Meet Images with Big Clusters and Features: A Cluster-weighted Multi-modal Co-clustering Method**
Chaoyang Zhang, Hang Xue, Kai Nie, Xihui Wu, Zhengzheng Lou, Shouyi Yang, Qinglei Zhou, Shizhe Hu (Corresponding author). [Information Processing and Management \(IPM\) 2024](#)
- **Joint Contrastive Triple-learning for Deep Multi-view Clustering**
Shizhe Hu, Guoliang Zou, Chaoyang Zhang, Zhengzheng Lou, Ruilin Geng, and Yangdong Ye, [Information Processing and Management \(IPM\) 2023](#), [ESI Highly-cited paper](#)
- **Deep Purified Feature Mining Model for Joint Named Entity Recognition and Relation Extraction**
Youwei Wang, Ying Wang, Zhongchuan Sun, Yinghao Li, Shizhe Hu, and Yangdong Ye. [Information Processing and Management \(IPM\) 2023](#)
- **Joint Specific and Correlated Information Exploration for Multi-view Action Clustering**
Shizhe Hu, Xiaoqiang Yan, Yangdong Ye, [Information Sciences 2020](#)
- **Multi-view Content-Context Information Bottleneck for Image Clustering**
Shizhe Hu, Bo Wang, Zhengzheng Lou, Yangdong Ye, [Expert Systems With Applications 2021](#)
- **DSPNet: Deep Scale Purifier Network for Dense Crowd Counting**
Xin Zeng, Yunpeng Wu, Shizhe Hu, Ruobin Wang and Yangdong Ye. [Expert Systems With Applications 2020](#)
- **Interactive Information Bottleneck for High-dimensional Co-occurrence Data Clustering**
Shizhe Hu, Ruobin Wang, and Yangdong Ye, [Applied Soft Computing 2021](#)
- **An Information Maximization Multi-task Clustering Method for Egocentric Temporal Segmentation**
Mingming Zhang, Xiaoqiang Yan, Shizhe Hu, Yangdong Ye, [Applied Soft Computing 2020](#)
- **Attentive Adversarial Collaborative Filtering**
Zhongchuan Sun, Bin Wu, Shizhe Hu, Mingming Zhang, and Yangdong Ye, [IEEE TSMC 2023](#)

- **Deep Multi-view Learning Methods: A Review**
Xiaoqiang Yan, [Shizhe Hu](#), Yiqiao Mao, Yangdong Ye, and Hui Yu, [Neurocomputing 2021](#) [ESI Highly-cited paper](#)
 - **Multi-task information bottleneck co-clustering for unsupervised cross-view human action categorization**
Xiaoqiang Yan, Zhengzheng Lou, [Shizhe Hu](#), Yangdong Ye, [ACM TKDD 2020](#)
 - **Learning a Deep Network with Cross-hierarchy Aggregation for Crowd Counting**
Qiang Guo, Xin Zeng, [Shizhe Hu](#), Sonephet Phoummixay and Yangdong Ye, [Knowledge-Based Systems 2021](#)
 - **Self-Supervised Temporal Autoencoder for Egocentric Action Segmentation**
Mingming Zhang, Dong liu, [Shizhe Hu](#), Xiaoqiang Yan, Zhongchuan Sun, and Yangdong Ye, [Engineering Applications of Artificial Intelligence 2023](#)
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Publications-Conferences

- **Calibrated Information Bottleneck for Trusted Multi-modal Clustering.**
[Shizhe Hu](#), Zhangwen Gou, Shuaiju Li, Jin Qin, Xiaoheng Jiang, Pei Lv, Mingliang Xu*, [ICLR 2026](#)
- **Diversity-oriented Deep Multi-modal Clustering.**
Yanzheng Wang#, Xin Yang#, Yujun Wang, [Shizhe Hu](#) (Co-Corresponding author), Mingliang Xu*, [NeurIPS 2025](#)
- **Structure-aware Granular-Ball based Information Bottleneck for Multi-modal Clustering.**
Zhengzheng Lou, Yuhan Zhan, Mingyang Lv, Yingxuan Li, Yuyang Du, [Shizhe Hu](#) (Corresponding author), [ICML 2026](#)
- **A Peer-review Look on Multi-modal Clustering: An Information Bottleneck Realization Method.**
Zhengzheng Lou, Hang Xue, Chaoyang Zhang, [Shizhe Hu](#) (Corresponding author), [ICML 2025](#)
- **Super Deep Contrastive Information Bottleneck for Multi-modal Clustering.**
Zhengzheng Lou, Ke Zhang, Yucong Wu, [Shizhe Hu](#) (Corresponding author), [ICML 2025](#)
- **Interest-driven Deep Multi-modal Clustering.**
Guoliang Zou#, Tongji Chen#, Sijia Li, Jin Qin, [Shizhe Hu](#)* (Co-Corresponding author), Yangdong Ye*, [AAAI 2026 Oral](#)
- **Multi-aspect Self-guided Deep Information Bottleneck for Multi-modal Clustering**
[Shizhe Hu](#), Jiahao Fan, Guoliang Zou, Yangdong Ye, [AAAI 2025](#)
- **Self-supervised Trusted Contrastive Multi-view Clustering with Uncertainty Refined**
[Shizhe Hu](#), Binyan Tian, Weibo Liu, Yangdong Ye. [AAAI 2025](#)
- **Proxy Information Bottleneck for Multi-modal Clustering**
Guoliang Zou, Sijia Li, Jinlan Wang, Tianyi Zhang, Pantongtong Li, Yangdong Ye*, [Shizhe Hu](#) (Co-Corresponding author). [ACM MM 2026](#)
- **Learning Dual Enhanced Representation for Contrastive Multi-view Clustering**
Guoliang Zou, Yangdong Ye, Tongji Chen, [Shizhe Hu](#) (Corresponding author). [ACM MM 2024](#)
- **A Parameter-free Multi-view Information Bottleneck Clustering Method by Cross-view Weighting**
[Shizhe Hu](#), Ruilin Geng, Zhaoxu Cheng, Chaoyang Zhang, Guoliang Zou, Zhengzheng Lou, Yangdong Ye, [ACM MM 2022](#)
- **Empowering Self-Balance of Deep Information Bottleneck for Multimodal Clustering**
Youwei Wang, Qiwei Miao, Haichuan Fang, Yangdong Ye, [Shizhe Hu](#) (Corresponding author). [IJCAI 2026](#)
- **Self-supervised Weighted Information Bottleneck for Multi-view Clustering**
Zhengzheng Lou, Chaoyang Zhang, Hang Xue, Yangdong Ye, Qinglei Zhou, [Shizhe Hu](#) (Corresponding author). [IJCAI 2024](#)
- **Content VS Context: How about "Walking Hand-In-Hand" for Image Clustering**
[Shizhe Hu](#), Zhenquan Hou, Zhengzheng Lou, Yangdong Ye, [IEEE ICASSP 2020](#)
- **Multi-task Clustering of Human Actions by Sharing Information**
Xiaoqiang Yan, [Shizhe Hu](#), Yangdong Ye, [IEEE CVPR 2017](#)
- **Heterogeneous Dual-Task Clustering with Visual-Textual Information**
Xiaoqiang Yan, Yiqiao Mao, [Shizhe Hu](#), Yangdong Ye, [SDM 2020](#)
- **Joint Contextual Transformer and Multi-scale Information Shared Network for Crowd Counting**
Xin Zeng, [Shizhe Hu](#), Huake Wang and Jinna Zhang, [PRAI 2022](#)